United States Patent	12409708
Kind Code	B2
Date of Patent	September 09, 2025
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Integrated device of pump and valve

Abstract

An integrated device of a pump and a valve is disclosed. The integrated device of a pump and a valve includes a drive unit, a valve assembly mounted on the drive unit and configured to be switchable to a first position or a second position by the drive unit, a pump mounted on the drive unit, and a controller configured to control operation of the drive unit and the pump. The valve assembly includes a plurality of inlets and a plurality of outlets. At the first position and the second position of the valve assembly, different fluid passages are formed in the valve assembly. The pump is configured to communicate with the valve assembly to allow a flow of fluid through the valve assembly.

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Appl. No.: **18/085124**

Filed: **December 20, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230398838 A1	Dec. 14, 2023

Foreign Application Priority Data

KR	10-2022-0070133	Jun. 09, 2022

Publication Classification

Int. Cl.: **B60H1/32** (20060101); **B60H1/00** (20060101)

U.S. Cl.:

CPC **B60H1/32** (20130101); **B60H1/00278** (20130101); B60H2001/00307 (20130101); B60H2001/3286 (20130101)

Field of Classification Search

CPC: B60H (1/00278); B60H (1/00485); B60H (1/32); B60H (1/32284); B60H (2001/00307); B60H (2001/3286)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) The present application claims priority to Korean Patent Application No. 10-2022-0070133, filed Jun. 9, 2022, the entire contents of which are incorporated herein by reference for all purposes.

TECHNICAL FIELD

(2) The present disclosure relates to an integrated device of a pump and a valve in which a pump and a valve are integrated with each other. More particularly, the present disclosure relates to an integrated device of a pump and a valve, and to a cooling water circulation system including the

same.
BACKGROUND
(3) An electric vehicle is a vehicle driven entirely or partially by a motor and is receiving a lot of attention due to its environmental friendliness. An electric vehicle includes a rechargeable battery that stores energy to be supplied to the motor.
(4) Since the performance of a battery can be greatly affected by temperature, an electric vehicle typically includes a temperature control system for the battery. The temperature control for the battery can be performed through an air cooling method, a water cooling method, or the like.
(5) Recently, as an increase in a driving distance on a full charge of an electric vehicle is required, the battery is becoming more sophisticated. Traditionally, the air cooling method was mainly used to control the temperature of the battery. However, as the battery becomes more sophisticated, there is a trend to change to a water cooling method.
SUMMARY
(6) An objective of the present disclosure is to provide an integrated device of a pump and a valve that can simplify the cooling loop of a battery.
(7) Another objective of the present disclosure is to provide an integrated device of a pump and a valve capable of reducing the costs by omitting parts.
(8) The objectives of the present disclosure are not limited to those mentioned above, and other objects not mentioned will be clearly understood by those skilled in the related art from the following description.
(9) In order to achieve the objectives of the present disclosure as described above and perform the characteristic functions of the present disclosure to be described later, the features of the present disclosure are as follows.
(10) According to some implementations described in the present disclosure, there is provided an integrated device of a pump and a valve. The integrated device of a pump and a valve includes a drive unit, a valve unit mounted on the drive unit and configured to switch between a first position and a second position by the drive unit, a pump unit mounted on the drive unit, and a controller configured to control operation of the drive unit and the pump unit. The valve unit includes a plurality of inlets and a plurality of outlets. At the first position and the second position of the valve unit, different fluid passages are formed in the valve unit. The pump unit is configured to communicate with the valve unit to allow a flow of fluid through the valve unit.
(11) According to some implementations described in the present disclosure, there is provided a cooling water circulation system. The cooling water circulation system includes a first cooling loop and a second cooling loop configured to circulate a coolant. The integrated device of a pump and a valve is disposed between the first and the second cooling loops and is switchable between the first position and the second position, thus forming different fluid passages between the first and the second cooling loops. The controller is configured to control the integrated device of a pump and a valve to be placed in the first or the second position based on a preset condition.
(12) According to the present disclosure, the integrated device of a pump and a valve enabling a simplified cooling loop configuration is provided.
(13) According to the present disclosure, the integrated device of a pump and a valve that provides weight and cost reduction effects is provided.
(14) The effects of the present disclosure are not limited to those mentioned above, and other effects not mentioned will be clearly understood by those skilled in the related art from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A shows a schematic diagram illustrating an example cooling mode for cooling a battery.

(2) FIG. **1**B shows a schematic diagram illustrating another example cooling mode for cooling the battery.

(3) FIG. **2** shows a perspective view of an example integrated device of a pump and a valve according to an implementation described in the present disclosure.

(4) FIG. **3** is an exploded perspective view of FIG. **2**.

(5) FIG. **4**A is a first direction sectional view of the example valve provided in the integrated device of a pump and a valve according to an implementation described in the present disclosure.

(6) FIG. **4**B is a second direction sectional view of the example integrated device of a pump and a valve, where the first direction and the second direction are perpendicular to each other.

(7) FIG. **4**C is a partially enlarged view of the example integrated device of a pump and a valve according to an implementation described in the present disclosure.

(8) FIG. **5**A is an exploded perspective view of an example controller part of the integrated device of a pump and a valve according to an implementation described in the present disclosure

(9) FIG. **5**B shows a bottom view of the example controller of the integrated device of a pump and a valve according to an implementation described in the present disclosure.

(10) FIG. **6**A to **6**C are views illustrating an example flow path according to an integrated cooling mode of the integrated device of a pump and a valve according to an implementation described in the present disclosure.

(11) FIG. **7**A to **7**C are views illustrating an example flow path according to another cooling mode of the integrated device of a pump and a valve according to an implementation described in the present disclosure.

DETAILED DESCRIPTION

(12) Specific structural or functional descriptions presented in the implementations of the present disclosure are only exemplified for the purpose of describing implementations according to the concept of the present disclosure, and the implementations according to the concept of the present disclosure may be implemented in various forms. In addition, it should not be construed as being limited to the implementations described herein, and should be understood to include all modifications, equivalents, and substitutes included in the spirit and scope of the present disclosure.

(13) Hereinafter, preferred implementations of the present disclosure will be described in detail with reference to the accompanying drawings.

(14) FIGS. **1**A and **1**B show an exemplary, water-cooled battery cooling system **500**. Cooling of a battery **502** in the cooling system **500** may be accomplished using a radiator **504** or a battery chiller **506**. To this end, an integrated cooling mode through the radiator **504** and a separate cooling mode through the battery chiller **506** using a three-way valve **508** and three T-shaped branch pipes **510***a*, **510***b*, **510***c* can be executed.

(15) FIG. **1**A shows an integrated cooling loop **530** for an integrated cooling mode. A coolant stored in a reservoir tank **512** starts circulation in the integrated cooling loop **530** by a water pump **514**. The coolant that has cooled the battery **502** exchanges heat in the radiator **504**. The coolant continues to circulate in the integrated cooling loop **530** and exchanges heat with the battery **502**.

(16) Two separate loops are configured in the separate cooling mode of FIG. **1**B. In a radiator loop **540**, the coolant circulates through the reservoir tank **512**, the water pump **514**, a water-cooled condenser **516** and the radiator **504**. The coolant in a chiller loop **550** is configured to circulate the battery **502** and the battery chiller **506** by a water pump **518**. Reference numeral **520** indicates a heater for increasing the temperature of the battery **502**. Here, the radiator loop **540** may be configured to cool other cooling subjects, such as power conversion parts.

(17) In order to enable the integrated cooling mode and the separate cooling mode as described

above, the three-way valve **508** and three T-shaped branch pipes **510***a*, **510***b*, **510***c* are used. In some vehicles, an integrated branch pipe in which three T-shaped branch pipes **510***a*, **510***b*, **510***c* are integrated into a single body is used.

(18) In this example cooling system **500**, the three-way valve **508**, three or more T-shaped branch pipes **510***a*, **510***b*, **510***c*, or an integrated branch pipe should be included. Also, the water pumps **514** and **518** should be provided separately. Therefore, a number of parts should be included, and the configuration of the flow path can be complicated.

(19) Accordingly, the present disclosure describes the integrated device of a pump and a valve in which a water pump and a valve are integrated and helps enable a simplified design of a flow path including the same. According to the present disclosure, since the three T-shaped branch pipes **510***a*, **510***b*, **510***c* or the integrated branch pipe can be omitted, the number of parts can be reduced and a connection path can be simplified, thereby reducing weight and cost, for instance.

(20) According to the integrated device **1** of a pump and a valve according to the present disclosure with reference to FIGS. **2** to **3**, a four-way valve is used instead of a three-way valve, and the four-way valve and one water pump are integrated. To this end, the integrated device **1** of a pump and a valve according to the present disclosure includes a valve unit **10**, a pump unit **20**, a drive unit **30**, and a controller **40**.

(21) The valve unit **10** includes two inlets **11**, **12**, one outlet **13** and a communication passage **14**. The coolant may be introduced into the valve unit **10** through the two inlets, the first inlet **11** and the second inlet **12**, The coolant may be discharged from the valve unit **10** through the outlet **13** and the communication passage **14**.

(22) The valve unit **10** includes a valve housing **16** in which a valve **15** is accommodated. The inlets **11**, **12**, the outlet **13** and the communication passage **14** may be formed into a single unit with the valve housing **16** or may be formed separately and detachably coupled to the valve housing **16** through bolts or the like.

(23) As shown in FIG. **4**A, the valve **15** forms a flow path for the coolant introduced through the inlets **11** and **12**. In some cooling modes, the valve **15** may communicate the first inlet **11** and the outlet **13** and the second inlet **12** and the communication passage **14** in communication. In addition, in some cooling modes, the valve **15** may be changed the position to communicate the first inlet **11** and the communication passage **14**, and to communicate the second inlet **12** and the outlet **13**. As a non-limiting example, the valve **15** may be a ball valve.

(24) The valve **15** can be positioned in a first position and a second position through rotation. In the first position, the first inlet **11** and the outlet **13** may communicate with each oipther. At the same time, the second inlet **12** and the communication passage **14** may communicate with each other. In the second position, the first inlet **11** and the communication passage **14** may communicate with each other. At the same time, the second inlet **12** and the outlet **13** may communicate with each other. To this end, the valve **15** includes four openings **115**, of which two openings **115** are configured to communicate with each other and the other two openings **115** to communicate with each other. In other words, two passages **215** are provided in the valve **15**.

(25) Referring to FIG. **4**B, the angles θ between the inlets **11** and **12**, the outlet **13** and the communication passage **14** are arranged to form obtuse angles therebetween. By the angles θ being obtuse angles, when the opening **115** of the valve **15** is aligned with the inlet **11** (or inlet **12**) and the outlet **13** (or the connection passage **14**), the coolant can flow along the two passages **215**. In addition, the two passages **215** extend curvedly in the valve **15**. The two passages **215** configured to be curved within the valve **15** can reduce the flow resistance that occurs when the coolant introduced through the inlets **11** and **12** is sharply bent toward the outlet **13** or the communication passage **14**. Since the inlets **11**, **12**, the outlet **13**, and the communication passage **14** are spaced apart from each other to form obtuse angles therebetween, and the two passages **215** curvedly extend in the valve **15**, the flow resistance of coolant introduced through the openings **115** can be reduced.

(26) As shown in FIG. **4**C, the valve housing **16** is provided with a sealing member **17**. The sealing member **17** prevents the coolant from leaking through the gap when the coolant flows. For instance, as shown in FIGS. **4**A and **4**B, a first seal **17***a* is disposed between the first inlet **11** and the valve **15**, a second seal **17***b* is disposed between the second inlet **12** and the valve **15**, a third seal **17***c* is disposed between the first outlet **11** and the valve **15**, and a fourth seal **17***d* is disposed between the second outlet **12** and the valve **15**. For example, when the first inlet **11** is aligned with one of the openings **115** of the valve **15**, the flowing coolant does not leak to the outside. Further, according to the present disclosure, the ring member **19** is mounted between each sealing member **17** and each inlet **11**, **12**, the outlet **13** and the communication passage **14**. The ring member **19** may provide an additional seal between elements. As a non-limiting example, the ring member **19** may be X-rings having an X-shaped cross-section. The ring member **19** may seal gaps formed between each groove **18** and the valve housing **16**, more specifically, a groove and inlets **11**, **12**, a groove and the outlet **13** and a groove and the communication passage **14**.
(27) The pump unit **20** is communicated to the valve unit **10**. In particular, the pump unit **20** is connected to the communication passage **14**, and the coolant flowing in through the valve unit **10** may be discharged through an outlet port **22** of the pump unit **20**.
(28) The drive unit **30** switches the position of the valve **15** of the valve unit **10** and allows the pump unit mounted on the drive unit **30** to operate. In some implementations, the drive unit **30** includes a holder portion **32** operably accommodating the pump unit **20**. In addition, the valve unit **10** is coupled to the drive unit **30**. In particular, the drive unit **30** is coupled to a shaft portion **315** of the valve **15** for switching the position of the valve **15**. Specifically, the drive unit **30** has a shaft connection portion **34** adjoining the holder portion **32** to rotatably connect the shaft portion **315**.
(29) Referring to FIGS. **5**A and **5**B, the controller **40** is configured to provide a control signal to the pump unit **20** and the drive unit **30**. According to an implementation described in the present disclosure, the controller **40** may be mounted on the drive unit **30**. As a non-limiting example, the controller **40** may be a printed circuit board mounted under the drive unit **30**.
(30) The controller **40** is electrically connected to the pump unit **20** and the drive unit **30**. In addition, the controller **40** is configured to control switching of the position of the valve unit **10** and the pumping of the pump unit **20** based on a required circulation rate of the coolant, a direction of the flow of the coolant, a flow rate of the coolant, a temperature of the coolant, and the like. For example, the controller **40** is configured to communicate with a flow sensor, a water temperature sensor, and the like, installed on a flow passage in which the integrated device **1** of a pump and a valve is disposed. The controller may control the pump unit **20** and the valve unit **10** based on the measured values collected from the sensors.
(31) The controller **40** may include a plurality of terminals **42***a*, **42***b*, **42***c* to be electrically connected to the drive unit **30**, a connector, and the pump unit **20**. According to the present disclosure, both the drive unit **30** and the pump unit **20** may be controlled by a single controller **40**.
(32) The controller **40** may rotate the valve **15** by controlling the angle of the drive unit **30** to change the direction of the coolant flow. When the integrated cooling mode or the separate cooling mode is required, the position of the valve **15** can be switched for each required mode. In addition, the controller **40** is configured to control the rotation number, e.g., rotations per minute (RPM), of the pump unit **20**. The controller **40** may operate the pump unit **20** at a certain RPM based on the required flow rate of coolant.
(33) A casing **50** is coupled to the drive unit **30**. In particular, the casing **50** is coupled to the drive unit **30** such that the controller **40** is encased and protected. The casing **50** may be coupled to the drive unit **30** with fastening members **60**, such as bolts.
(34) The integrated device **1** of a pump and a valve according to the present disclosure may further include a bracket **70**. The bracket **70** holds the integrated device **1** of a pump and a valve and allows it to be mounted on a mounting target. In some implementations of the present disclosure, a pair of brackets **70***a*, **70***b* facing each other may be coupled to the integrated device **1** to surround

the drive unit **30**.
(35) Hereinafter, the operation of the integrated device **1** of a pump and a valve according to the present disclosure will be described. FIGS. **6**A to **6**C and FIGS. **7**A to **7**C show the integrated cooling mode and the separate cooling mode, respectively.
(36) As depicted in FIGS. **6**A to **6**C, the integrated device **1** of a pump and a valve according to the present disclosure may cool the battery **502** by the integrated cooling mode. When the integrated cooling mode is requested, the controller **40** commands the drive unit **30** to position the valve **15** in the first position. In the first position of the valve **15**, the first inlet **11** and the outlet **13** communicate with each other (P**1**-P**2**). At the same time, the second inlet **12** and the discharge port **22** communicate with each other, i.e., through the connection passage **14** (P**3**-P**4**). In addition, the controller **40** operates the pump unit **20** at a predetermined RPM so that the coolant exchanged heat with the radiator **504** cools the battery **502**.
(37) As shown in FIGS. **7**A to **7**C, the integrated device **1** of a pump and a valve can cool the battery **502** in a separate cooling mode. In the separate cooling mode, two separate cooling loops are configured. In the separate cooling mode, the first inlet **11** communicates with the discharge port **22** (P**1**-P**4**), and the second inlet **12** communicates with the outlet **13** (P**2**-P**3**). To this end, the controller **40** rotates the drive unit **30** so that the valve **15** is positioned in the second position.
(38) According to the present disclosure, it is possible to reduce the cost and compact the package by eliminating the T-shaped branch pipe, hose, and the like. In addition, the present disclosure may reduce the hydraulic power loss through the simplification of the flow passage.
(39) Although the present disclosure was described with reference to specific implementations shown in the drawings, it is apparent to those skilled in the art that the present disclosure may be changed and modified in various ways without departing from the scope of the present disclosure, which is described in the following claims.

Claims

1. A cooling device comprising: a motor; a valve unit configured to switch between a first position and a second position by the motor, the valve unit having a plurality of inlets and a plurality of outlets, wherein different fluid passages are formed in the valve unit in the first position and the second position, respectively; a pump in fluid communication with the valve unit and configured to circulate a fluid through the valve unit; and a controller configured to control the motor and the pump, wherein the motor comprises a holder portion configured to accommodate the pump therein.
2. The cooling device of claim 1, wherein the plurality of inlets comprises a first inlet and a second inlet, and the plurality of outlets comprises a first outlet and a second outlet, wherein in the first position, (i) a fluid passage is formed between the first inlet and the first outlet, and (ii) a fluid passage is formed between the second inlet and the second outlet, and wherein in the second position, (i) a fluid passage is formed between the first inlet and the second outlet, and (i) a fluid passage is formed between the second inlet and the first outlet.
3. The cooling device of claim 2, wherein the fluid passing through the second outlet is discharged through the pump.
4. The cooling device of claim 3, wherein the second outlet comprises a communication passage and a discharge port, wherein the communication passage is configured to provide fluid communication between the valve unit and the pump, and wherein the discharge port is in fluid communication with the communication passage and configured to discharge the fluid passing through the pump.
5. The cooling device of claim 2, wherein the first inlet, the second inlet, the first outlet, and the second outlet are arranged in the valve unit to form obtuse angles with each other.
6. The cooling device of claim 1, wherein the valve unit comprises: a valve housing; and a valve rotatably provided inside the valve housing to rotate between the first position and the second

position.
7. The cooling device of claim 6, wherein the valve comprises a first passage and a second passage, and wherein the first passage and the second passage have curved shapes within the valve.
8. The cooling device of claim 6, wherein the valve unit further comprises: a shaft portion extending from the valve, wherein the shaft portion is connected to a shaft connection portion of the motor that is configured to rotate the shaft portion to switch a position of the valve.
9. The cooling device of claim 6, wherein the plurality of inlets comprises a first inlet and a second inlet, and the plurality of outlets comprises a first outlet and a second outlet, wherein a plurality of seals are mounted in the valve housing, the plurality of seals comprising first to fourth seals, and wherein the first seal is disposed between the first inlet and the valve, the second seal is disposed between the second inlet and the valve, the third seal is disposed between the first outlet and the valve, and the fourth seal is disposed between the second outlet and the valve.
10. The cooling device of claim 9, wherein ring members are interposed between the first seal and the first inlet, between the second seal and the second inlet, between the third seal and the first outlet, and between the fourth seal and the second outlet, respectively.
11. The cooling device of claim 10, wherein each of the ring members has an X-shaped cross section.
12. The cooling device of claim 1, further comprising a bracket mounted to the motor to surround the motor.
13. The cooling device of claim 1, wherein the controller is mounted to the motor.
14. The cooling device of claim 13, further comprising a casing mounted to the motor and configured to protect the controller.
15. The cooling device of claim 1, wherein the controller comprises a plurality of terminals configured to provide connection with the motor and the pump.
16. A cooling system comprising: a first cooling loop and a second cooling loop configured to circulate a coolant; an integrated device having a pump and a valve, the integrated device being disposed between the first and second cooling loops and being switchable between a first position and a second position that form different fluid passages between the first and second cooling loops; and a controller configured to control the integrated device to be placed in the first position or the second position based on a preset condition, wherein the preset condition includes one or more of a flow rate and a temperature of the coolant.
17. The cooling system of claim 16, wherein in the first position, the first cooling loop and the second cooling loop communicate with each other via the integrated device, and wherein in the second position, the first cooling loop and the second cooling loop are separated from each other via the integrated device.
18. A cooling system comprising: a first cooling loop and a second cooling loop configured to circulate a coolant; an integrated device having a pump and a valve, the integrated device being disposed between the first and second cooling loops and being switchable between a first position and a second position that form different fluid passages between the first and second cooling loops; and a controller configured to control the integrated device to be placed in the first position or the second position based on a preset condition, wherein the coolant circulating in the first cooling loop is in a heat exchange relationship with a radiator and power conversion parts of a vehicle, and the coolant circulating in the second cooling loop is in a heat exchange relationship with a battery of the vehicle.